Guitar Exercises - Section VIII, Level 67

A harmonic minor interval sequences, arpeggio overlays, mode functions, and melodic application

FIXED KEY: A | 10 MODULES | 792 TONES

Level 66 connected five positions. Level 67 now reorganizes those notes by interval, chord, mode function, and progression. The purpose is not to memorize seven new scale boxes; it is to hear why the same A-harmonic-minor pitch collection behaves differently over i, iiø, III+, iv, V7, VI, and vii°7.

1. Hear interval structure

Thirds, fourths, fifths, and sixths reveal the scale geometry more clearly than another consecutive scale run.

2. Overlay the harmony

Triad and seventh-chord arpeggios show which tones are structural inside each mode and progression.

3. Assign mode function

Each mode is tied to its diatonic chord. The chord decides which degree should sound like a target and which should remain color.

A harmonic minor: pitch, harmony, and characteristic intervals

```
Scale:      A  B  C  D  E  F  G# A
Degrees:    1  2  b3 4  5  b6 7  1

Diatonic triads:   Am  Bdim  Caug  Dm  E   F   G#dim
Seventh chords:    AmMaj7  Bm7b5  Cmaj7#5  Dm7  E7  Fmaj7  G#dim7

Characteristic motions:
F -> G#       augmented second: b6 -> 7
G# -> A       leading tone: 7 -> 1
D -> G#       augmented fourth inside D Dorian #4
C -> G#       augmented fifth inside C Ionian #5
```

Written workload

Module	Tones
A - intervals	240
B - triads	56
C - sevenths	56
D - modes	112
E - functions	56
F - five cadences	80
G - progressions	48
H - Mode 5	40
I - resolutions	40
J - capstone	64
TOTAL	792

Session protocol

Choose one interval pattern, one arpeggio overlay, and one functional progression. Keep the Level-66 fingering and alternate-picking orientation unless the exercise explicitly changes the system. Sing the target degree before the final note of every cell.

MODE RULE Name the chord first, then the mode. Do not use the mode name as a substitute for hearing the harmony.

Guitar Exercises - Section VIII, Level 67

Harmonic-minor mode and chord-function atlas

7 MODES | 7 CHORDS | ONE PITCH COLLECTION

MODES ARE ROTATIONS OF A HARMONIC MINOR; THE CHORD CHANGES THE TONAL CENTER

Mode	Name	Formula	Chord function	Characteristic color
Mode 1	A harmonic minor	1 2 b3 4 5 b6 7 1	AmMaj7 / i	7->1 and b6->7
Mode 2	B Locrian natural 6	1 b2 b3 4 b5 6 b7 1	Bm7b5 / iiø7	natural 6 above a Locrian tonic
Mode 3	C Ionian #5	1 2 3 4 #5 6 7 1	Cmaj7#5 / III+	#5 over a major tonic
Mode 4	D Dorian #4	1 2 b3 #4 5 6 b7 1	Dm7 / iv7	#4 over a minor iv chord
Mode 5	E Phrygian dominant	1 b2 3 4 5 b6 b7 1	E7 / V7	b2 plus major 3 over V7
Mode 6	F Lydian #2	1 #2 3 #4 5 6 7 1	Fmaj7 / VImaj7	#2 and #4 over VImaj7
Mode 7	G# ultralocrian	1 b2 b3 b4 b5 b6 bb7 1	G#dim7 / vii°7	b2, b4, b5, bb7 over vii°7

Primary tonic and predominant functions

Mode 1 supplies AmMaj7/i. Mode 2 supplies Bm7b5/iiø. Mode 4 supplies Dm7/iv. Mode 6 supplies Fmaj7/VI. These colors support the minor-key center.

Dominant and unstable functions

Mode 5 supplies E7/V7. Mode 7 supplies G#dim7/vii°7. Mode 3 supplies Cmaj7#5/III+, whose augmented fifth points strongly toward A-minor or E7 material.

Function test

Play the same seven notes beginning on A, E, and G#. If A still sounds like the only possible tonic, sustain the local chord beneath the mode and stop each phrase on the chord root. The mode is functioning only when the new local center is audible.

Guitar Exercises - Section VIII, Level 67

Module A - P1 tonic frame interval sequences

4 PATTERNS | 48 TONES | FRETS 5-10

PAIR THE SCALE DEGREES; DO NOT INSERT THE INTERVENING NOTES

Each twelve-tone cell contains six diatonic interval pairs. The interval name counts scale letters, not a fixed number of semitones. Harmonic minor therefore produces major, minor, augmented, and diminished interval qualities inside the same sequence.

Diatonic thirds

Pairs: A-C | B-D | C-E | D-F | E-G# | F-A

```
Nte|A---C---B---D---C---E---D---F---E---G#--F---A---
Deg|1---b3--2---4---b3--5---4---b6--5---7---b6--1---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-------------------------------------------------
  D|-----------------------------------6-------7---
  A|-----------5-------7---5---8---7-------8-------
  E|5---8---7-------8-----------------------------
```

Diatonic fourths

Pairs: A-D | B-E | C-F | D-G# | E-A | F-B

```
Nte|A---D---B---E---C---F---D---G#--E---A---F---B---
Deg|1---4---2---5---b3--b6--4---7---5---1---b6--2---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-------------------------------------------------
  D|---------------------------6-------7-------9---
  A|----5-------7-------8---5-------7-------8-------
  E|5-------7-------8-----------------------------
```

Diatonic fifths

Pairs: A-E | B-F | C-G# | D-A | E-B | F-C

```
Nte|A---E---B---F---C---G#--D---A---E---B---F---C---
Deg|1---5---2---b6--b3--7---4---1---5---2---b6--b3--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-----------------------------------------5---
  D|-------------------6-------7-------9-----------
  A|----7-------8-----------5-------7-------8-------
  E|5-------7-------8-----------------------------
```

Diatonic sixths

Pairs: A-F | B-G# | C-A | D-B | E-C | F-D

```
Nte|A---F---B---G#--C---A---D---B---E---C---F---D---
Deg|1---b6--2---7---b3--1---4---2---5---b3--b6--4---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-----------------------------------5-------7---
  D|-----------6-------7-------9-------------------
  A|----8-----------------5-------7-------8-------
  E|5-------7-------8-----------------------------
```

Guitar Exercises - Section VIII, Level 67

Module A - P2 second-degree frame interval sequences

4 PATTERNS | 48 TONES | FRETS 7-12

PAIR THE SCALE DEGREES; DO NOT INSERT THE INTERVENING NOTES

Each twelve-tone cell contains six diatonic interval pairs. The interval name counts scale letters, not a fixed number of semitones. Harmonic minor therefore produces major, minor, augmented, and diminished interval qualities inside the same sequence.

Diatonic thirds

Pairs: B-D | C-E | D-F | E-G# | F-A | G#-B

```
Nte|B---D---C---E---D---F---E---G#--F---A---G#--B---
Deg|2---4---b3--5---4---b6--5---7---b6--1---7---2---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|-----------------------------------7-------9---
  A|-----------7-------8---7---11--8-------11------
  E|7---10--8-------10----------------------------
```

Diatonic fourths

Pairs: B-E | C-F | D-G# | E-A | F-B | G#-C

```
Nte|B---E---C---F---D---G#--E---A---F---B---G#--C---
Deg|2---5---b3--b6--4---7---5---1---b6--2---7---b3--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|---------------------------7-------9-------10--
  A|----7-------8-------11--7-------8-------11------
  E|7-------8-------10----------------------------
```

Diatonic fifths

Pairs: B-F | C-G# | D-A | E-B | F-C | G#-D

```
Nte|B---F---C---G#--D---A---E---B---F---C---G#--D---
Deg|2---b6--b3--7---4---1---5---2---b6--b3--7---4---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-------------------------------------------7---
  D|-------------------7-------9-------10----------
  A|----8-------11----------7-------8-------11------
  E|7-------8-------10----------------------------
```

Diatonic sixths

Pairs: B-G# | C-A | D-B | E-C | F-D | G#-E

```
Nte|B---G#--C---A---D---B---E---C---F---D---G#--E---
Deg|2---7---b3--1---4---2---5---b3--b6--4---7---5---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------7-------9---
  D|-----------7-------9-------10------------------
  A|----11----------------7-------8-------11------
  E|7-------8-------10----------------------------
```

Guitar Exercises - Section VIII, Level 67

Module A - P3 flat-third frame interval sequences

4 PATTERNS | 48 TONES | FRETS 8-13

PAIR THE SCALE DEGREES; DO NOT INSERT THE INTERVENING NOTES

Each twelve-tone cell contains six diatonic interval pairs. The interval name counts scale letters, not a fixed number of semitones. Harmonic minor therefore produces major, minor, augmented, and diminished interval qualities inside the same sequence.

Diatonic thirds

Pairs: C-E | D-F | E-G# | F-A | G#-B | A-C

```
Nte|C---E---D---F---E---G#--F---A---G#--B---A---C---
Deg|b3--5---4---b6--5---7---b6--1---7---2---1---b3--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|-----------------------------------9-------10--
  A|-----------8-------11--8---12--11------12------
  E|8---12--10------12----------------------------
```

Diatonic fourths

Pairs: C-F | D-G# | E-A | F-B | G#-C | A-D

```
Nte|C---F---D---G#--E---A---F---B---G#--C---A---D---
Deg|b3--b6--4---7---5---1---b6--2---7---b3--1---4---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|---------------------------9-------10------12--
  A|----8-------11------12--8-------11------12------
  E|8-------10------12----------------------------
```

Diatonic fifths

Pairs: C-G# | D-A | E-B | F-C | G#-D | A-E

```
Nte|C---G#--D---A---E---B---F---C---G#--D---A---E---
Deg|b3--7---4---1---5---2---b6--b3--7---4---1---5---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-------------------------------------------9---
  D|-------------------9-------10------12----------
  A|----11------12----------8-------11------12------
  E|8-------10------12----------------------------
```

Diatonic sixths

Pairs: C-A | D-B | E-C | F-D | G#-E | A-F

```
Nte|C---A---D---B---E---C---F---D---G#--E---A---F---
Deg|b3--1---4---2---5---b3--b6--4---7---5---1---b6--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------9-------10--
  D|------------9-------10------12-----------------
  A|----12----------------8-------11------12------
  E|8-------10------12----------------------------
```

Guitar Exercises - Section VIII, Level 67

Module A - P4 fourth-degree frame interval sequences

4 PATTERNS | 48 TONES | FRETS 10-16

PAIR THE SCALE DEGREES; DO NOT INSERT THE INTERVENING NOTES

Each twelve-tone cell contains six diatonic interval pairs. The interval name counts scale letters, not a fixed number of semitones. Harmonic minor therefore produces major, minor, augmented, and diminished interval qualities inside the same sequence.

Diatonic thirds

Pairs: D-F | E-G# | F-A | G#-B | A-C | B-D

```
Nte|D---F---E---G#--F---A---G#--B---A---C---B---D---
Deg|4---b6--5---7---b6--1---7---2---1---b3--2---4---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|-----------------------------------10------12--
  A|-----------11------12--11--14--12------14------
  E|10--13--12------13----------------------------
```

Diatonic fourths

Pairs: D-G# | E-A | F-B | G#-C | A-D | B-E

```
Nte|D---G#--E---A---F---B---G#--C---A---D---B---E---
Deg|4---7---5---1---b6--2---7---b3--1---4---2---5---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-----------------------------------------------
  D|---------------------------10------12------14--
  A|----11------12------14--11------12------14------
  E|10------12------13----------------------------
```

Diatonic fifths

Pairs: D-A | E-B | F-C | G#-D | A-E | B-F

```
Nte|D---A---E---B---F---C---G#--D---A---E---B---F---
Deg|4---1---5---2---b6--b3--7---4---1---5---2---b6--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|-------------------------------------------10--
  D|-------------------10------12------14----------
  A|----12------14----------11------12------14------
  E|10------12------13----------------------------
```

Diatonic sixths

Pairs: D-B | E-C | F-D | G#-E | A-F | B-G#

```
Nte|D---B---E---C---F---D---G#--E---A---F---B---G#--
Deg|4---2---5---b3--b6--4---7---5---1---b6--2---7---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-----------------------------------------------
  B|-----------------------------------------------
  G|---------------------------------10------13--
  D|-----------10------12------14-----------------
  A|----14----------------11------12------14------
  E|10------12------13----------------------------
```

Standard tuning. Fixed key: A harmonic minor. Mode names describe harmonic function, not seven unrelated box systems.

Guitar Exercises - Section VIII, Level 67

Module A - P5 fifth-degree frame interval sequences

4 PATTERNS | 48 TONES | FRETS 12-17

PAIR THE SCALE DEGREES; DO NOT INSERT THE INTERVENING NOTES

Each twelve-tone cell contains six diatonic interval pairs. The interval name counts scale letters, not a fixed number of semitones. Harmonic minor therefore produces major, minor, augmented, and diminished interval qualities inside the same sequence.

Diatonic thirds

Pairs: E-G# | F-A | G#-B | A-C | B-D | C-E

```
Nte|E---G#--F---A---G#--B---A---C---B---D---C---E---
Deg|5---7---b6--1---7---2---1---b3--2---4---b3--5---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-------------------------------------------------
  D|-----------------------------------12------14--
  A|-----------12------14--12--15--14------15------
  E|12--16--13------16----------------------------
```

Diatonic fourths

Pairs: E-A | F-B | G#-C | A-D | B-E | C-F

```
Nte|E---A---F---B---G#--C---A---D---B---E---C---F---
Deg|5---1---b6--2---7---b3--1---4---2---5---b3--b6--
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-------------------------------------------------
  D|---------------------------12------14------15--
  A|----12------14------15--12------14------15------
  E|12------13------16----------------------------
```

Diatonic fifths

Pairs: E-B | F-C | G#-D | A-E | B-F | C-G#

```
Nte|E---B---F---C---G#--D---A---E---B---F---C---G#--
Deg|5---2---b6--b3--7---4---1---5---2---b6--b3--7---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|-----------------------------------------13--
  D|-------------------12------14------15----------
  A|----14------15----------12------14------15------
  E|12------13------16----------------------------
```

Diatonic sixths

Pairs: E-C | F-D | G#-E | A-F | B-G# | C-A

```
Nte|E---C---F---D---G#--E---A---F---B---G#--C---A---
Deg|5---b3--b6--4---7---5---1---b6--2---7---b3--1---
Pair|1---1---2---2---3---3---4---4---5---5---6---6---
  e|-------------------------------------------------
  B|-------------------------------------------------
  G|---------------------------------13------14--
  D|-----------12------14------15-----------------
  A|----15----------------12------14------15------
  E|12------13------16----------------------------
```

Standard tuning. Fixed key: A harmonic minor. Mode names describe harmonic function, not seven unrelated box systems.

Guitar Exercises - Section VIII, Level 67

Module B - harmonized-triad arpeggio overlays

7 TRIADS | 56 TONES

ROOT-THIRD-FIFTH-OCTAVE | REPEAT THE TOP | EXACT REVERSE

The scale is reorganized into its seven diatonic triads. The augmented III chord and diminished ii and vii chords are structural results of the raised seventh, not optional chromatic decorations.

i - Am

8-tone arpeggio mirror

```
Nte|A----C----E----A----A----E----C----A----
Deg|1----b3---5----1----1----5----b3---1----
  D|---------------7----7------------------
  A|----------7-------------7--------------
  E|5----8------------------------8----5----
```

ii° - Bdim

8-tone arpeggio mirror

```
Nte|B----D----F----B----B----F----D----B----
Deg|2----4----b6---2----2----b6---4----2----
  D|---------------9----9------------------
  A|-----5----8-------------8----5---------
  E|7-----------------------------------7----
```

III+ - Caug

8-tone arpeggio mirror

```
Nte|C----E----G#---C----C----G#---E----C----
Deg|b3---5----7----b3---b3---7----5----b3---
  G|---------------5----5------------------
  D|----------6-------------6--------------
  A|-----7------------------------7---------
  E|8-----------------------------------8----
```

iv - Dm

8-tone arpeggio mirror

```
Nte|D----F----A----D----D----A----F----D----
Deg|4----b6---1----4----4----1----b6---4----
  G|---------------7----7------------------
  D|----------7-------------7--------------
  A|5----8------------------------8----5----
```

V - E

8-tone arpeggio mirror

```
Nte|E----G#---B----E----E----B----G#---E----
Deg|5----7----2----5----5----2----7----5----
  G|---------------9----9------------------
  D|-----6----9-------------9----6---------
  A|7-----------------------------------7----
```

VI - F

8-tone arpeggio mirror

```
Nte|F----A----C----F----F----C----A----F----
Deg|b6---1----b3---b6---b6---b3---1----b6---
  B|---------------6----6------------------
  G|----------5-------------5--------------
  D|-----7------------------------7---------
  A|8-----------------------------------8----
```

vii° - G#dim

8-tone arpeggio mirror

```
Nte|G#---B----D----G#---G#---D----B----G#---
Deg|7----2----4----7----7----4----2----7----
  B|---------------9----9------------------
  G|----------7-------------7--------------
  D|6----9------------------------9----6----
```

Guitar Exercises - Section VIII, Level 67

Module C - harmonized seventh-chord arpeggio overlays

7 SEVENTH CHORDS | 56 TONES

THE SEVENTH REVEALS THE MODE FUNCTION MORE CLEARLY THAN THE TRIAD ALONE

Play the four chord tones ascending and descending. Accent the quality-defining tones: 7 in AmMaj7, b5 in Bm7b5, #5 in Cmaj7#5, 3 in E7, and the diminished seventh F in G#dim7.

i(maj7) - AmMaj7

4 up + 4 down

```
Nte|A----C----E----G#---G#---E----C----A----
Deg|1----b3---5----7----7----5----b3---1----
  D|---------------6----6-------------------
  A|----------7-------------7--------------
  E|5----8------------------------8----5----
```

iiø7 - Bm7b5

4 up + 4 down

```
Nte|B----D----F----A----A----F----D----B----
Deg|2----4----b6---1----1----b6---4----2----
  D|---------------7----7-------------------
  A|-----5----8-------------8----5---------
  E|7-----------------------------------7----
```

IIImaj7#5 - Cmaj7#5

4 up + 4 down

```
Nte|C----E----G#---B----B----G#---E----C----
Deg|b3---5----7----2----2----7----5----b3---
  D|----------6----9----9----6--------------
  A|-----7------------------------7---------
  E|8-----------------------------------8----
```

iv7 - Dm7

4 up + 4 down

```
Nte|D----F----A----C----C----A----F----D----
Deg|4----b6---1----b3---b3---1----b6---4----
  G|--------------5----5-------------------
  D|----------7--------------7--------------
  A|5----8------------------------8----5----
```

V7 - E7

4 up + 4 down

```
Nte|E----G#---B----D----D----B----G#---E----
Deg|5----7----2----4----4----2----7----5----
  G|---------------7----7-------------------
  D|-----6----9--------------9----6---------
  A|7-----------------------------------7----
```

VImaj7 - Fmaj7

4 up + 4 down

```
Nte|F----A----C----E----E----C----A----F----
Deg|b6---1----b3---5----5----b3---1----b6---
  B|--------------5----5-------------------
  G|----------5--------------5--------------
  D|-----7------------------------7---------
  A|8-----------------------------------8----
```

vii°7 - G#dim7

4 up + 4 down

```
Nte|G#---B----D----F----F----D----B----G#---
Deg|7----2----4----b6---b6---4----2----7----
  B|---------------6----6-------------------
  G|----------7--------------7--------------
  D|6----9------------------------9----6----
```

Guitar Exercises - Section VIII, Level 67

Module D - one-octave mode mirrors

2 MODES | 32 TONES

ASCEND ONE OCTAVE, RE-ATTACK THE ROOT, AND DESCEND THE SAME ROUTE

All routes are rotations of the P1 pitch map. Speak the mode-relative formula, not the A-harmonic-minor degree names. Sustain the local chord or drone so the starting note becomes a real tonal center.

Mode 1: A harmonic minor - AmMaj7 / i

Mode-relative formula: 1 2 b3 4 5 b6 7 1

```
Nte|A---B---C---D---E---F---G#--A---A---G#--F---E---D---C---B---A---
Deg|1---2---b3--4---5---b6--7---1---1---7---b6--5---4---b3--2---1---
  e|-------------------------------------------------------------------
  B|-------------------------------------------------------------------
  G|-------------------------------------------------------------------
  D|-----------------------6---7---7---6-------------------------------
  A|-----------5---7---8-------------------8---7---5-------------------
  E|5---7---8-----------------------------------------8---7---5---
```

Mode 2: B Locrian natural 6 - Bm7b5 / iiø7

Mode-relative formula: 1 b2 b3 4 b5 6 b7 1

```
Nte|B---C---D---E---F---G#--A---B---B---A---G#--F---E---D---C---B---
Deg|1---b2--b3--4---b5--6---b7--1---1---b7--6---b5--4---b3--b2--1---
  e|-------------------------------------------------------------------
  B|-------------------------------------------------------------------
  G|-------------------------------------------------------------------
  D|-------------------6---7---9---9---7---6-----------------------
  A|--------5---7---8---------------------------8---7---5-----------
  E|7---8-------------------------------------------------8---7---
```

Guitar Exercises - Section VIII, Level 67

Module D - one-octave mode mirrors

2 MODES | 32 TONES

ASCEND ONE OCTAVE, RE-ATTACK THE ROOT, AND DESCEND THE SAME ROUTE

All routes are rotations of the P1 pitch map. Speak the mode-relative formula, not the A-harmonic-minor degree names. Sustain the local chord or drone so the starting note becomes a real tonal center.

Mode 3: C Ionian #5 - Cmaj7#5 / III+

Mode-relative formula: 1 2 3 4 #5 6 7 1

```
Nte|C---D---E---F---G#--A---B---C---C---B---A---G#--F---E---D---C---
Deg|1---2---3---4---#5--6---7---1---1---7---6---#5--4---3---2---1---
  e|-------------------------------------------------------------------
  B|-------------------------------------------------------------------
  G|---------------------------5---5---------------------------------
  D|---------------6---7---9-----------9---7---6---------------------
  A|----5---7---8-----------------------------------8---7---5-------
  E|8-------------------------------------------------------8---
```

Mode 4: D Dorian #4 - Dm7 / iv7

Mode-relative formula: 1 2 b3 #4 5 6 b7 1

```
Nte|D---E---F---G#--A---B---C---D---D---C---B---A---G#--F---E---D---
Deg|1---2---b3--#4--5---6---b7--1---1---b7--6---5---#4--b3--2---1---
  e|-------------------------------------------------------------------
  B|-------------------------------------------------------------------
  G|------------------------5---7---7---5---------------------------
  D|------------6---7---9-------------------9---7---6---------------
  A|5---7---8-----------------------------------------8---7---5---
  E|-------------------------------------------------------------------
```

Guitar Exercises - Section VIII, Level 67

Module D - one-octave mode mirrors

3 MODES | 48 TONES

ASCEND ONE OCTAVE, RE-ATTACK THE ROOT, AND DESCEND THE SAME ROUTE

All routes are rotations of the P1 pitch map. Speak the mode-relative formula, not the A-harmonic-minor degree names. Sustain the local chord or drone so the starting note becomes a real tonal center.

Mode 5: E Phrygian dominant

E7 / V7 | 1 b2 3 4 5 b6 b7 1

```
Nte|E--F--G#-A--B--C--D--E--E--D--C--B--A--G#-F--E--
Deg|1--b2-3--4--5--b6-b7-1--1--b7-b6-5--4--3--b2-1--
  G|--------------5--7--9--9--7--5-----------------
  D|------6--7--9-------------------9--7--6--------
  A|7--8-----------------------------------8--7--
```

Mode 6: F Lydian #2

Fmaj7 / VImaj7 | 1 #2 3 #4 5 6 7 1

```
Nte|F--G#-A--B--C--D--E--F--F--E--D--C--B--A--G#-F--
Deg|1--#2-3--#4-5--6--7--1--1--7--6--5--#4-3--#2-1--
  B|--------------------6--6----------------------
  G|------------5--7--9--------9--7--5--------------
  D|---6--7--9-------------------------9--7--6-----
  A|8-----------------------------------------8--
```

Mode 7: G# ultralocrian

G#dim7 / vii°7 | 1 b2 b3 b4 b5 b6 bb7 1

```
Nte|G#-A--B--C--D--E--F--G#-G#-F--E--D--C--B--A--G#-
Deg|1--b2-b3-b4-b5-b6-bb71--1--bb7b6-b5-b4-b3-b2-1--
  B|-----------------6--9--9--6-------------------
  G|---------5--7--9--------------9--7--5----------
  D|6--7--9-----------------------------9--7--6--
```

Mode 5, 6, and 7 comparison

Mode	Root chord	Stable chord tones	Characteristic tensions
E Phrygian dominant	E7	E G# B D	F(b2), C(b6)
F Lydian #2	Fmaj7	F A C E	G#(#2), B(#4)
G# ultralocrian	G#dim7	G# B D F	A(b2), C(b4), E(b6)

Guitar Exercises - Section VIII, Level 67

Module E - mode-function melodic cells

7 FUNCTIONS | 56 TONES

BEGIN ON A CHORD TONE, TOUCH THE CHARACTERISTIC MODE COLOR, AND RETURN TO THE CHORD

These eight-tone phrases connect the mode name to its diatonic chord. A correct function cell remains recognizable if the characteristic color is removed; the chord tones provide the structure.

Mode 1 - AmMaj7

Chord tones frame the characteristic mode color.

```
Nte|A----C----E----G#---F----E----C----A----
Deg|1----b3---5----7----b6---5----b3---1----
  D|---------------6-----------------------
  A|----------7---------8----7-------------
  E|5----8-----------------------8----5----
```

Mode 2 - Bm7b5

Chord tones frame the characteristic mode color.

```
Nte|B----D----F----A----G#---F----D----B----
Deg|2----4----b6---1----7----b6---4----2----
  D|---------------7----6-------------------
  A|-----5----8-------------8----5---------
  E|7-----------------------------------7----
```

Mode 3 - Cmaj7#5

Chord tones frame the characteristic mode color.

```
Nte|C----E----G#---B----A----G#---E----C----
Deg|b3---5----7----2----1----7----5----b3---
  D|----------6----9----7----6--------------
  A|-----7-----------------------7---------
  E|8-----------------------------------8----
```

Mode 4 - Dm7

Chord tones frame the characteristic mode color.

```
Nte|D----F----A----C----G#---A----F----D----
Deg|4----b6---1----b3---7----1----b6---4----
  G|--------------5-----------------------
  D|----------7---------6----7--------------
  A|5----8-----------------------8----5----
```

Mode 5 - E7

Chord tones frame the characteristic mode color.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|5----b6---7----2----4----b3---2----5----
  G|-------------------7----5--------------
  D|----------6----9-------------9---------
  A|7----8-----------------------------7----
```

Mode 6 - Fmaj7

Chord tones frame the characteristic mode color.

```
Nte|F----G#---A----C----E----D----C----F----
Deg|b6---7----1----b3---5----4----b3---b6---
  B|-------------------5-------------------
  G|---------------5---------7----5---------
  D|-----6----7-------------------------------
  A|8-----------------------------------8----
```

Mode 7 - G#dim7

Chord tones frame the characteristic mode color.

```
Nte|G#---B----D----F----E----D----B----G#---
Deg|7----2----4----b6---5----4----2----7----
  B|---------------6----5-------------------
  G|----------7--------------7--------------
  D|6----9-----------------------9----6----
```

Guitar Exercises - Section VIII, Level 67

Module F - five-position i-iv-V7-i melodic applications

3 PHRASES | 48 TONES

| Am | Dm | E7 | Am | - FOUR QUARTER NOTES PER BAR

The phrase begins on a chord tone in every measure. E7 includes G#, and the final Am measure restores A as tonic. Convert notes 2-4 into shuffle subdivisions only after the quarter-note version is secure.

P1 - TONIC FRAME cadence phrase

Bar targets: Am root/triad | Dm root/triad | E7 with G# | Am resolution

```
Nte|A---C---E---A---|D---F---A---D---|E---G#--B---D---|A---C---E---A---|
Deg|1---b3--5---1---|4---b6--1---4---|5---7---2---4---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|------------10--|
  G|----------------|------------7---|----------------|----5---9-------|
  D|------------7---|----------------|----6-----------|7---------------|
  A|--------7-------|5---8-----------|7-----------5---|----------------|
  E|5---8-----------|--------5-------|--------7-------|----------------|
```

P2 - SECOND-DEGREE FRAME cadence phrase

Bar targets: Am root/triad | Dm root/triad | E7 with G# | Am resolution

```
Nte|A---C---E---A---|D---F---A---D---|E---G#--B---D---|A---C---E---A---|
Deg|1---b3--5---1---|4---b6--1---4---|5---7---2---4---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|----------------|
  B|------------10--|----------------|----------------|10----------10--|
  G|----------------|------------7---|----------------|--------9-------|
  D|7---------------|--------7-------|----------------|----10----------|
  A|--------7-------|----8-----------|7---11----------|----------------|
  E|----8-----------|10--------------|--------7---10--|----------------|
```

P3 - FLAT-THIRD FRAME cadence phrase

Bar targets: Am root/triad | Dm root/triad | E7 with G# | Am resolution

```
Nte|A---C---E---A---|D---F---A---D---|E---G#--B---D---|A---C---E---A---|
Deg|1---b3--5---1---|4---b6--1---4---|5---7---2---4---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|----------------|
  B|------------10--|----------------|----------------|10----------10--|
  G|----------------|----------------|----------------|--------9-------|
  D|----------------|------------12--|--------9-------|----10----------|
  A|12--------------|----8---12------|----11----------|----------------|
  E|----8---12------|10--------------|12----------10--|----------------|
```

Guitar Exercises - Section VIII, Level 67

Module F - five-position i-iv-V7-i melodic applications

2 PHRASES | 32 TONES

| Am | Dm | E7 | Am | - FOUR QUARTER NOTES PER BAR

The phrase begins on a chord tone in every measure. E7 includes G#, and the final Am measure restores A as tonic. Convert notes 2-4 into shuffle subdivisions only after the quarter-note version is secure.

P4 - FOURTH-DEGREE FRAME cadence phrase

Bar targets: Am root/triad | Dm root/triad | E7 with G# | Am resolution

```
Nte|A---C---E---A---|D---F---A---D---|E---G#--B---D---|A---C---E---A---|
Deg|1---b3--5---1---|4---b6--1---4---|5---7---2---4---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|----13----------|
  G|------------14--|----------------|----------------|14----------14--|
  D|----10----------|------------12--|----------------|--------14------|
  A|12--------------|--------12------|----11--14------|----------------|
  E|--------12------|10--13----------|12----------10--|----------------|
```

P5 - FIFTH-DEGREE FRAME cadence phrase

Bar targets: Am root/triad | Dm root/triad | E7 with G# | Am resolution

```
Nte|A---C---E---A---|D---F---A---D---|E---G#--B---D---|A---C---E---A---|
Deg|1---b3--5---1---|4---b6--1---4---|5---7---2---4---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|------------17--|
  B|----------------|------------15--|----------------|----13----------|
  G|------------14--|----------------|----------------|14--------------|
  D|----------------|12--------------|------------12--|--------14------|
  A|12--15----------|--------12------|--------14------|----------------|
  E|--------12------|----13----------|12--16----------|----------------|
```

Variation: preserve beat-one targets and rewrite only notes 2-4 of each bar in the neighboring position.

Guitar Exercises - Section VIII, Level 67

Module G - progression-specific melodic applications

3 PROGRESSIONS | 48 TONES

THE HARMONIC-MINOR CHORD FAMILY EXTENDS BEYOND i-iv-V7-i

Each progression emphasizes a different harmonic-minor function: tonic expansion, functional predominant-dominant motion, or augmented-major contrast. Keep one chord tone on beat 1 of every bar.

i-VI-iv-V7

Four chord-tone events per measure.

```
Nte|A---C---E---A---|F---A---C---F---|D---F---A---D---|E---G#--B---D---|
Deg|1---b3--5---1---|b6--1---b3--b6--|4---b6--1---4---|5---7---2---4---|
Chd|>Am-Am--Am--Am--|>F--F---F---F---|>Dm-Dm--Dm--Dm--|>E7-E7--E7--E7--|
  e|----------------|----------------|----------------|----------------|
  B|------------10--|----------------|----------------|----------------|
  G|----------------|------------10--|----------------|----------------|
  D|----------------|----------------|------------12--|--------9-------|
  A|12--------------|8---12----------|----8---12------|----11----------|
  E|----8---12------|--------8-------|10--------------|12----------10--|
```

ii°-V7-i-i

Four chord-tone events per measure.

```
Nte|B---D---F---B---|E---G#--B---D---|A---C---E---A---|C---E---A---A---|
Deg|2---4---b6--2---|5---7---2---4---|1---b3--5---1---|b3--5---1---1---|
Chd|>BdiBdimBdimBdim|>E7-E7--E7--E7--|>Am-Am--Am--Am--|>Am-Am--Am--Am--|
  e|----------------|----------------|----------------|----------------|
  B|------------12--|----------------|------------10--|--------10------|
  G|----------------|----------------|----------------|----9-----------|
  D|9---------------|--------9-------|----------------|10--------------|
  A|--------8-------|----11----------|12--------------|------------12--|
  E|----10----------|12----------10--|----8---12------|----------------|
```

III+-VI-ii°-V7

Four chord-tone events per measure.

```
Nte|C---E---G#--C---|F---A---C---F---|B---D---F---B---|E---G#--B---D---|
Deg|b3--5---7---b3--|b6--1---b3--b6--|2---4---b6--2---|5---7---2---4---|
Chd|>CauCaugCaugCaug|>F--F---F---F---|>BdiBdimBdimBdim|>E7-E7--E7--E7--|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|------------12--|----------------|
  G|----------------|------------10--|----------------|----------------|
  D|------------10--|----------------|9---------------|--------9-------|
  A|--------11------|8---12----------|--------8-------|----11----------|
  E|8---12----------|--------8-------|----10----------|12----------10--|
```

Guitar Exercises - Section VIII, Level 67

Module H - E Phrygian dominant through five positions

5 CELLS | 40 TONES

E-F-G#-B-D-C-B-E | b2 AND b6 COLOR A CLEAR E7 TARGET

Use these phrases only while E7 or a dominant E pedal is active. The major third G# must remain clear. F and C create the Phrygian-dominant color but should not obscure the E7 chord tones.

P1 - E Phrygian dominant

E7 function: target 1, 3, 5, or b7.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|1----b2---3----5----b7---b6---5----1----
  G|-----------------------------------9----
  D|----------6-------------------9---------
  A|7----8-------------5-------------------
  E|--------------7---------8--------------
```

P2 - E Phrygian dominant

E7 function: target 1, 3, 5, or b7.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|1----b2---3----5----b7---b6---5----1----
  G|-----------------------------------9----
  D|----------------------------9---------
  A|7----8----11-------------------------
  E|--------------7----10---8-------------
```

P3 - E Phrygian dominant

E7 function: target 1, 3, 5, or b7.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|1----b2---3----5----b7---b6---5----1----
  B|-------------------------------12-------
  G|-----------------------------------9----
  D|---------------9----------------------
  A|-----8----11---------------------------
  E|12-----------------10---8-------------
```

P4 - E Phrygian dominant

E7 function: target 1, 3, 5, or b7.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|1----b2---3----5----b7---b6---5----1----
  B|-------------------------------12-------
  D|------------------------10--------14---
  A|----------11---14----------------------
  E|12---13------------10-----------------
```

P5 - E Phrygian dominant

E7 function: target 1, 3, 5, or b7.

```
Nte|E----F----G#---B----D----C----B----E----
Deg|1----b2---3----5----b7---b6---5----1----
  G|-----------------------------16--------
  D|-------------------12-------------14---
  A|---------------14--------15-------------
  E|12---13---16---------------------------
```

Dominant release

After every cell, play G#-A and stop. This proves that the E7 mode belongs to the larger A-minor cadence rather than becoming an unrelated E-centered scale exercise.

Guitar Exercises - Section VIII, Level 67

Module I - characteristic chord overlays and resolutions

5 CELLS | 40 TONES

THE UNUSUAL CHORDS BECOME MUSICAL WHEN THEIR VOICES RESOLVE

Each eight-tone cell states one harmonic-minor seventh chord and then moves to a destination chord. Listen for the leading tone, altered fifth, diminished fifth, or diminished seventh resolving by half step or c

iMaj7 -> i

Four-tone source chord -> four-tone destination.

```
Nte|A----C----E----G#---A----E----C----A----
Deg|1----b3---5----7----1----5----b3---1----
  D|--------------6----7-------------------
  A|----------7-------------7--------------
  E|5----8-----------------------8----5----
```

IIImaj7#5 -> i

Four-tone source chord -> four-tone destination.

```
Nte|C----E----G#---B----A----E----C----A----
Deg|b3---5----7----2----1----5----b3---1----
  D|----------6----9----7-------------------
  A|-----7-------------------7--------------
  E|8---------------------------8----5----
```

iiø7 -> V7

Four-tone source chord -> four-tone destination.

```
Nte|B----D----F----A----G#---B----D----E----
Deg|2----4----b6---1----7----2----4----5----
  G|-----------------------------7----9----
  D|--------------7----6----9--------------
  A|-----5----8-----------------------------
  E|7-----------------------------------
```

vii°7 -> i

Four-tone source chord -> four-tone destination.

```
Nte|G#---B----D----F----A----C----E----A----
Deg|7----2----4----b6---1----b3---5----1----
  B|--------------6-------------5----10---
  G|----------7-------------5-------------
  D|6----9--------------7-------------------
```

VImaj7 -> V7

Four-tone source chord -> four-tone destination.

```
Nte|F----A----C----E----G#---B----D----E----
Deg|b6---1----b3---5----7----2----4----5----
  B|--------------5-----------------------
  G|----------5-------------------7----9----
  D|-----7-------------6----9--------------
  A|8--------------------------------------
```

Voice-leading audit

Name the smallest motion between corresponding chord tones. Then replay the cell while sustaining common tones and minimizing every moving voice. The fretboard shape is secondary to the voice-leading result.

Guitar Exercises - Section VIII, Level 67

Module J - sixteen-bar mode-function and progression capstone

16 BARS | 64 TONES

FOUR SYSTEMS OF FOUR BARS | THE MODE LABEL FOLLOWS THE ACTIVE CHORD

The capstone rotates through all seven harmonic-minor modes inside one coherent A-minor form. Every bar contains four quarter notes. Later passes may add one articulation without changing the beat-one target or mode function.

Bars 1-4: AmMaj7 | Dm7 | E7 | Am

```
Nte|A---C---E---G#--|D---F---A---C---|E---F---G#--B---|A---C---E---A---|
Deg|1---b3--5---7---|4---b6--1---b3--|5---b6--7---2---|1---b3--5---1---|
Chd|>AmMAmMaAmMaAmMa|>Dm7Dm7-Dm7-Dm7-|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
Mod|Mode------------|Mode------------|Mode------------|Mode------------|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|----------------|
  G|----------------|----------------|----------------|----------------|
  D|-----------6----|----------------|--------6-------|-----------7----|
  A|--------7-------|5---8-----------|7---8-----------|--------7-------|
  E|5---8-----------|--------5---8---|-----------7----|5---8-----------|
```

Bars 5-8: Cmaj7#5 | Fmaj7 | Bm7b5 | E7

```
Nte|C---E---G#--B---|F---G#--A---C---|B---D---F---A---|E---F---G#--D---|
Deg|b3--5---7---2---|b6--7---1---b3--|2---4---b6--1---|5---b6--7---4---|
Chd|>CmaCmajCmajCmaj|>FmaFmajFmajFmaj|>Bm7Bm7bBm7bBm7b|>E7-E7--E7--E7--|
Mod|Mode------------|Mode------------|Mode------------|Mode------------|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|----------------|
  G|----------------|----------------|----------------|----------------|
  D|-----------9----|----------------|9---------------|----------------|
  A|--------11------|8---11--12------|--------8---12--|----8---11------|
  E|8---12----------|-----------8---|----10----------|12----------10--|
```

Bars 9-12: Dm7 | G#dim7 | AmMaj7 | E7

```
Nte|D---G#--A---F---|G#--B---D---F---|A---C---E---G#--|E---G#--B---D---|
Deg|4---7---1---b6--|7---2---4---b6--|1---b3--5---7---|5---7---2---4---|
Chd|>Dm7Dm7-Dm7-Dm7-|>G#dG#diG#diG#di|>AmMAmMaAmMaAmMa|>E7-E7--E7--E7--|
Mod|Mode------------|Mode------------|Mode------------|Mode------------|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|----------------|
  G|----------------|----------------|----------------|----------------|
  D|----------------|----------------|----10----------|----------------|
  A|----11--12------|11--14----------|12----------11--|----11--14------|
  E|10----------13--|--------10--13--|--------12------|12----------10--|
```

Bars 13-16: Am | Fmaj7 | E7 | Am

```
Nte|A---B---C---E---|F---G#--A---E---|E---F---G#--B---|A---C---E---A---|
Deg|1---2---b3--5---|b6--7---1---5---|5---b6--7---2---|1---b3--5---1---|
Chd|>Am-Am--Am--Am--|>FmaFmajFmajFmaj|>E7-E7--E7--E7--|>Am-Am--Am--Am--|
Mod|Mode------------|Mode------------|Mode------------|Mode------------|
  e|----------------|----------------|----------------|----------------|
  B|----------------|----------------|----------------|----------------|
  G|----------------|----------------|----------------|-----------14--|
  D|----------------|----------------|----------------|----------------|
  A|12--14--15------|--------12------|-----------14--|12--15----------|
  E|-----------12--|13--16------12--|12--13--16------|--------12------|
```

Standard tuning. Fixed key: A harmonic minor. Mode names describe harmonic function, not seven unrelated box systems.

Guitar Exercises - Section VIII, Level 67

Practice progression, error diagnosis, mastery gate, and log

ADVANCE BY FUNCTIONAL HEARING, NOT MODE-NAME RECALL

THE MODE IS CORRECT ONLY WHEN THE CHORD AND RESOLUTION ARE AUDIBLE

Ten-phase practice progression

- [ ] 1. P1 thirds, fourths, fifths, and sixths.
- [ ] 2. Translate each interval type through P2-P5.
- [ ] 3. Seven triad overlays.
- [ ] 4. Seven seventh-chord overlays.
- [ ] 5. Seven mode mirrors over sustained chords.
- [ ] 6. Seven mode-function cells.
- [ ] 7. Five i-iv-V7-i phrases.
- [ ] 8. Three alternate progression families.
- [ ] 9. Five E Phrygian dominant cells.
- [ ] 10. Characteristic resolutions and capstone.

Error diagnosis

MODE NAME WITHOUT FUNCTION	hold the chord or drone and stop on its root
INTERVAL RUN BECOMES SCALE	remove the connecting notes; play only the pair
AUG2 RUSH	isolate F-G# and keep equal subdivisions
V7 SOUNDS MINOR	accent G# before adding F or C color
DIMINISHED CELL SOUNDS RANDOM	state G#dim7, then resolve every voice
III+ SOUNDS UNSTABLE	land C-E-G# first; add B only after the triad is clear
FORM LOSS	return to quarter-note chord tones and name each bar

Mastery gate before Level 68

- [ ] Play all twenty interval cells and identify the diatonic interval family by ear.
- [ ] Spell and arpeggiate all seven triads and seven seventh chords.
- [ ] Perform all seven mode mirrors while the local chord remains audible.
- [ ] Play all seven function cells and explain the characteristic color.
- [ ] Complete five i-iv-V7-i phrases and all three alternate progression families.
- [ ] Perform E Phrygian dominant in five registers and resolve every cell to A minor.
- [ ] Complete all five characteristic chord-resolution cells and the sixteen-bar capstone.

Practice log

Date	Module / mode	Chord function	D/U	BPM	Clean?	Weak interval / resolution	Next target

NEXT Level 68: E Phrygian dominant, diminished-seventh symmetry, altered dominant colors, and harmonic-minor improvisation synthesis.

Standard tuning. Fixed key: A harmonic minor. Mode names describe harmonic function, not seven unrelated box systems.